Unit no. ° 7

ONE-VARIABLE DESCRIPTIVE STATISTICS

7-18 December 2026

I Counts, frequencies and mean

Definitions

A series describes a characteristic of a population. If the value x_i appears n_i times, n_i is its frequency. The non-zero total frequency is $N = \sum_{i=1}^{p} n_i$; its relative frequency is $f_i = n_i/N$. The mean is

$$\overline{x} = \frac{\sum_{i=1}^{p} n_i x_i}{N} = \sum_{i=1}^{p} f_i x_i.$$

The range is $x_{\max} - x_{\min}$.

Example 7.1: Shoe sizes · booklet data

We study the shoe sizes of 30 pupils:

36	42	40	41	41	38	39	43	42	42	42	40	36	40	39
41	40	38	44	42	40	42	43	38	43	36	41	41	44	43

Complete the counts, calculate the frequencies, the mean and the range.

Shoe size	36	37	38	39	40	41	42	43	44
Count	**3**	**0**	**3**	**2**	**5**	**5**	**6**	**4**	**2**

Solution:

Counts: 3 , 0, 3, 2, 5, 5, 6, 4, 2, with sum 30. The frequencies are these counts divided by 30, i.e. as a percentage 10; 0; 10; approximately 6 ,67 ; 16,67 ; 16,67 ; 20 ; 13,33 ; 6,67. $\sum n_i x_i$ = 1217, therefore $\overline{x} = 1217/30 \approx 40{,}57$. The range is 8. The mean is not necessarily a value in the series. The sum of rounded percentages may differ slightly from 100.

II Median and cumulative frequencies

Definition and convention

A median is a number such that at least half the values are less than or equal to it, and at least half greater than or equal to it. In the ordered series of N values, if N is odd, take the index $(N+1)/2$; if N is even, take the mean of the values with indices $N/2$ and $N/2+1$.

Watch out. With repeated values, more than 50% of the values may be less than or equal to the median.

Example 7.2

Complete the cumulative frequencies of the shoe sizes and determine the median.

Shoe size	36	37	38	39	40	41	42	43	44
Count	3	0	3	2	5	5	6	4	2
ICC	**3**	**3**	**6**	**8**	**13**	**18**	**24**	**28**	**30**

Solution:

The middle ranks are 15 and 16. The cumulative frequencies go from 13 for 40 to 18 for 41: the two middle values are 41, so M_e = 41. At least half the shoe sizes are less than or equal to 41, and at least half greater than or equal to 41.

Example 7.3

Calculate the medians of 1; 2; 2; 8; 9 and of 1; 2; 8; 9.

Solution:

The third value of the first data set is 2. In the second, (2 + 8) /2 = 5; the chosen median is not a value of the data set.

III Quartiles and box plot

Definitions

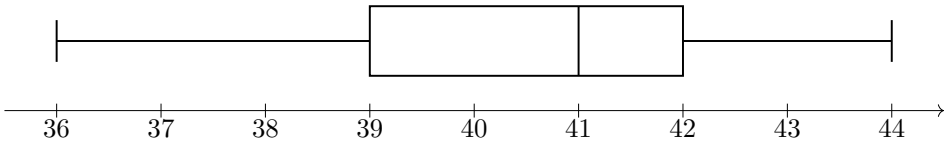
The first quartile Q_1 is the value with index $\lceil N/4 \rceil$: the smallest value in the series such that at least 25% of the values are less than or equal to it. Q_3 is the value with index $\lceil 3N/4 \rceil$. The IQ range is $Q_3 - Q_1$. The box plot shows the minimum, Q_1, median, Q_3, maximum.

Example 7.4

Determine Q_1, Q_3 and the interquartile range of the shoe sizes, then draw the box plot.

Solution:

$N/4 = 7{,}5$: position 8, so $Q_1 = 39$. $3N/4 = 22{,}5$: position 23, so $Q_3 = 42$. Interquartile range 3. The diagram uses (36; 39; 41; 42; 44).



Example 7.5

Determine the quartiles of 1; 2; 2; 3; 4; 4; 4; 5. Are exactly 25% of the values less than or equal to Q_1 ?

Solution:

$Q_1 = 2$ (2nd value), $Q_3 = 4$ (6th value). Here $3/8 = 37{,}5$ % of the values are less than or equal to Q_1.We must say "at least".

IV Adding or removing a value

Properties

Adding a to a series of size N gives $\overline{x}' = (N\overline{x} + a)/(N+1)$. Removing one occurrence of a, for $N > 1$, gives $\overline{x}' = (N\overline{x} - a)/(N-1)$. Use the exact sum, without intermediate rounding. Re-examine the median in the new ordered series.

Example 7.6

In the series of shoe sizes, the sum is 1217 and the median 41.

1. Add a size 36 shoe and calculate the new summary statistics.

2. Start again from the original series, remove a size 44 and repeat.

Solution:

1. Sum 1253, count 31: $\overline{x}' = 1253/31 \approx 40{,}42$. The middle position 16 still gives 41. 2. Sum 1173, count 29: $\overline{x}'' = 1173/29 \approx 40{,}45$. The middle position 15 gives 41. The mean decreases in both cases; the median remains unchanged. The situations are independent.

Example 7.7

Justify that adding the mean does not change the mean. Can the range decrease when a value is added?

Solution:

If $a = \overline{x}$, $(N\overline{x} + \overline{x})/(N+1) = \overline{x}$. When a value is added, the minimum can only stay the same or decrease, the maximum can only stay the same or increase: the range does not decrease. When a value is removed, it does not increase.

V Variance and standard deviation

Definitions

$V = \dfrac{\sum n_i(x_i - \overline{x})^2}{N}$ is the variance; $\sigma = \sqrt{V}$ is the standard deviation. It measures the spread around the mean and has the same unit as the data. We can also calculate $V = \dfrac{\sum n_i x_i^2}{N} - \overline{x}^2$.

Example 7.8

Calculate the variance and standard deviation of the shoe sizes.

Solution:

$\sum n_i x_i^2 = 49519$, therefore

$$V = \frac{49519}{30} - \left(\frac{1217}{30}\right)^2 = \frac{4481}{900} \approx 4{,}979.$$

$\sigma = \sqrt{4481}/30 \approx 2{,}23$. The standard deviation is not the maximum distance from the mean.

Example 7.9

Compare A : 8; 10; 12 and B : 0; 10; 20 with the pairs mean-standard deviation and median-interquartile range.

Solution:

The means and medians are all 10. $V_A = 8/3$, $\sigma_A \approx 1{,}63$; $V_B = 200/3$, $\sigma_B \approx 8{,}16$. Quartiles : 8, 12 for A ; 0, 20 for B. Interquartile ranges 4 and 20. B is much more spread out despite identical centres.

Guidance for the teacher

The descriptive variance divides by N, without the correction by $N - 1$. Also check the quartile convention in software.

VI Linearity of the mean

Property

Replace each x_i by $kx_i + c$ gives a mean $k\overline{x} + c$, a standard deviation $|k|\sigma$ and a variance k^2V. Adding a constant changes neither the variance nor the standard deviation.

Example 7.10

Initial mean 12 /20, standard deviation 3. Calculate the new statistics for three independent changes, without capping the marks: add 1 ,5; multiply by 0 ,8; multiply by 1 ,1 then add 0,5.

Solution:

Means: 13 ,5, 9,6, 13,7. Standard deviations: 3, 2,4, 3,3. Multiplying the marks by 1 ,1 increases them; this does not mean making the assessment harder. With a cap at 20, the transformation would no longer be affine on all the data.

```
from math import sqrt

def moyenne_ecart_type(valeurs):
    n = len(valeurs)
    if n == 0:
        raise ValueError("liste vide")
    m = sum(valeurs)/n
    v = sum((x-m)**2 for x in valeurs)/n
    return m, sqrt(v)
```

Example 7.11

Explain the program; predict its result for `[8, 10, 12]`.

Solution:

It checks that the list is non-empty, computes the mean, the mean of the squared deviations, then its square root. Result: (10; $\sqrt{8/3}$) $\approx (10; 1{,}633)$.

VII Continuous data grouped into classes

Definitions and loss of information

A continuous quantitative variable can take all the values of an interval. Here, the observations are grouped into classes of equal width; the midpoint of $[a; b[$ equals $(a+b)/2$. If the means m_i of the classes are known, the exact overall mean is $\sum n_i m_i / N$. Otherwise, replacing them with the class midpoints gives an estimate justified by a uniform model within each class.

Example 7.12: Heights · booklet data

Complete, estimate the mean from the class midpoints and determine the median class.

Class (cm)	$[150; 160[$	$[160; 170[$	$[170; 180[$	$[180; 190[$	$[190; 200[$
Count	8	22	35	28	7
Frequency (%)	**8**	**22**	**35**	**28**	**7**
Centre (cm)	**155**	**165**	**175**	**185**	**195**
ICC	**8**	**30**	**65**	**93**	**100**

Solution:

Class width 10 cm, total count 100. Frequencies as percentages: 8, 22, 35, 28, 7. Centres: 155, 165, 175, 185, 195 cm. The weighted sum of the class midpoints is 17540, so the estimated mean is 175,4 cm. The middle positions 50, 51 lie in [170; 180[, the median class.

Example 7.13

If the exact class means were 154; 166; 174; 184; 196 cm, what would the overall mean be?

Solution:

$(8 \times 154 + 22 \times 166 + 35 \times 174 + 28 \times 184 + 7 \times 196)/100 = 174{,}98$ cm.

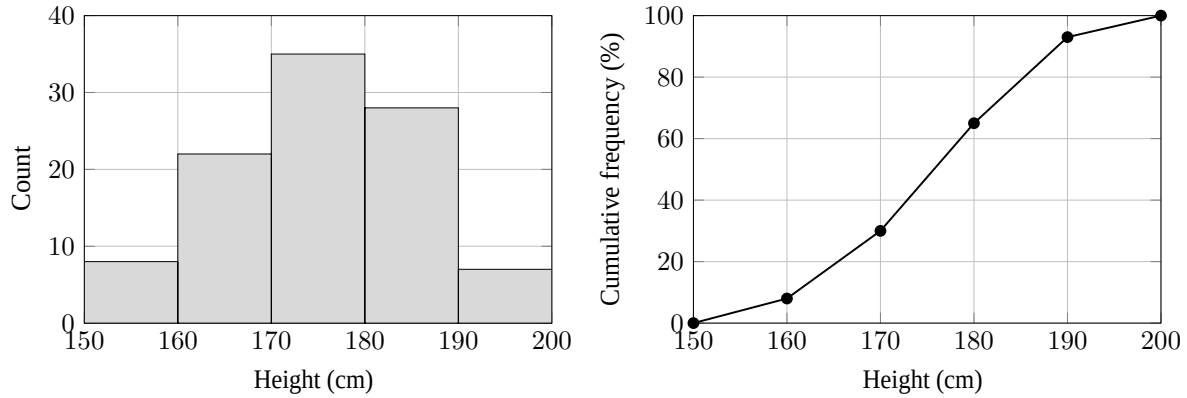
VIII Histogram and cumulative frequencies

Representations

The area of each rectangle in a histogram is proportional to the frequency. For equal class widths, the heights can be the frequencies. The cumulative frequency polygon joins the points (upper bound; cumulative frequency), starting at the first lower bound with 0.

Example 7.14

Draw the two graphs for the heights in the previous example.



Estimate the median

Assuming a uniform distribution in [170; 180[, the cumulative frequency increases linearly from 30% to 65%. Find the x-value corresponding to 50%.

Example 7.15

Calculate this estimate and state its scope.

Solution:

We solve $(m - 170)/10 = (50 - 30)/(65 - 30)$, hence $m = 170 + 10 \times 20/35 \approx 175{,}71$ cm. This is an estimate under the assumption of uniformity, not the exact median of the individual data.
The class frequencies alone do not determine the latter.